CHAPTER 6 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Risk Analysis, Attack Detection, and Cyber–Physical Monitoring Techniques

Companion figures, risk records, a sensor-consistency example, and a detector-evaluation exercise

Learning goal: connect monitoring evidence to validated detection performance and mission risk.

AI-based detection: models and evidence

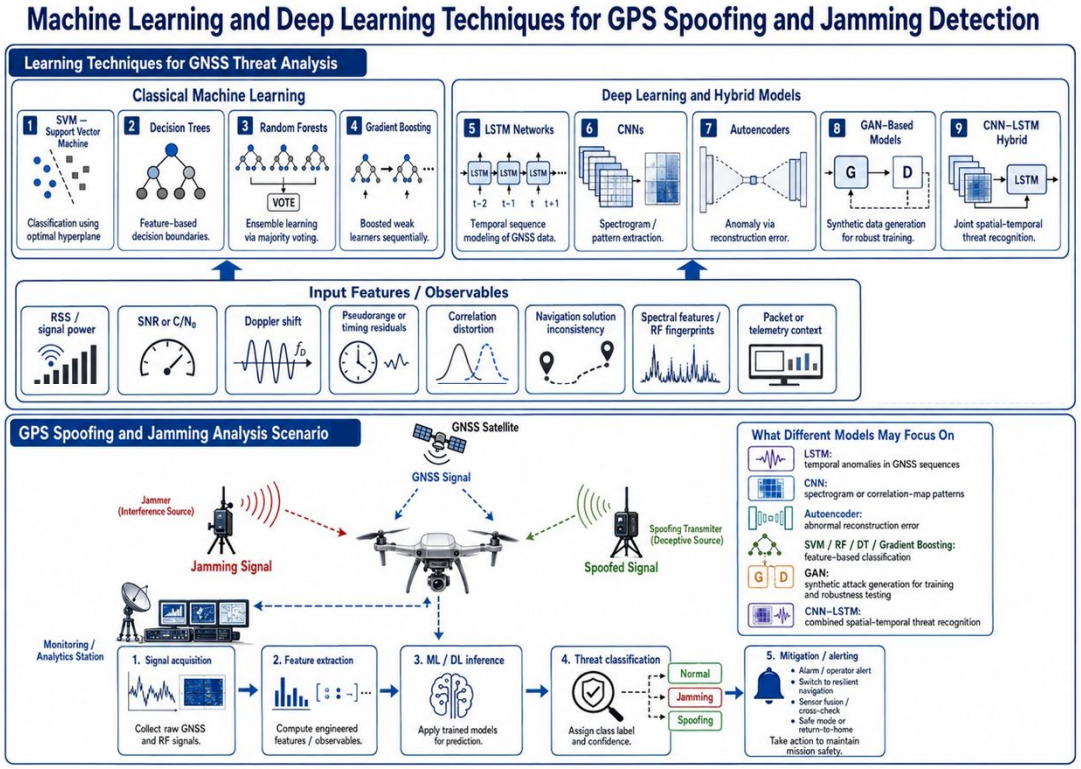


Figure S6.1. Feature selection and model choice depend on available signals and test conditions. An anomaly score or class label requires validation before it supports a response decision.

Conceptual overview. Listed methods and outcomes do not establish measured detection performance.

Telemetry and flight-dynamics anomalies

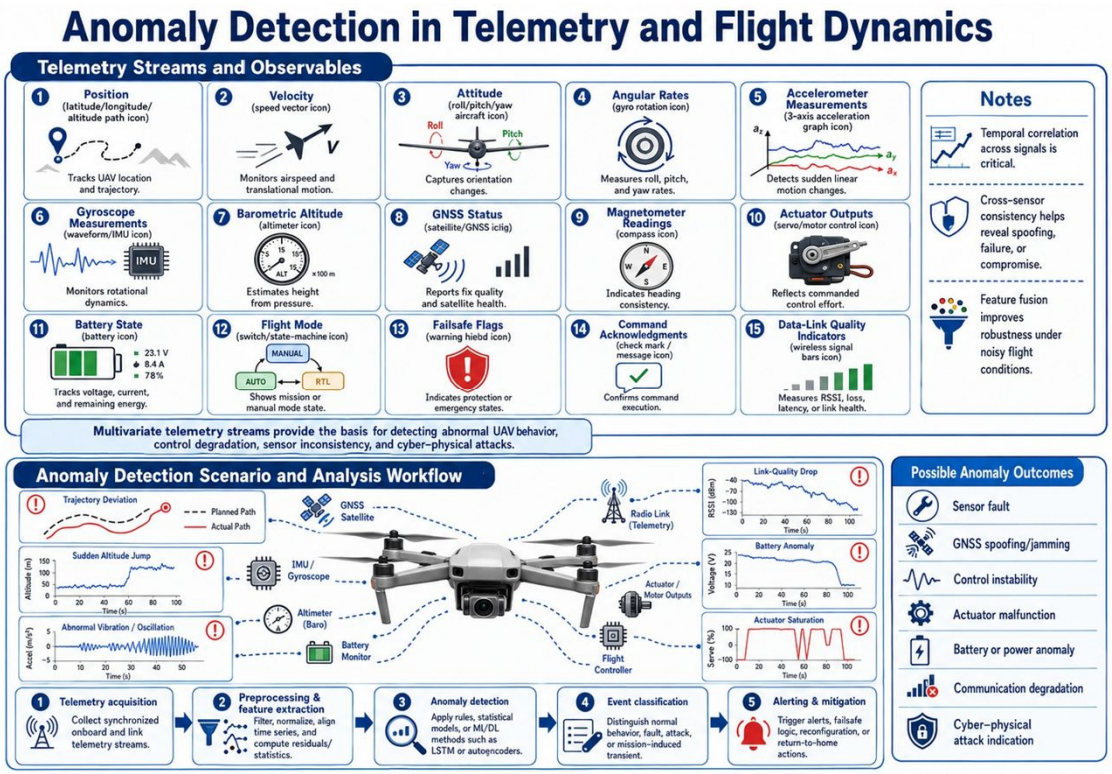


Figure S6.2. Telemetry monitoring combines timing, vehicle dynamics, and operating context. Faults, disturbances, stale data, and attacks can produce overlapping symptoms.

Conceptual overview. Listed methods and outcomes do not establish measured detection performance.

Sensor fusion and consistency monitoring

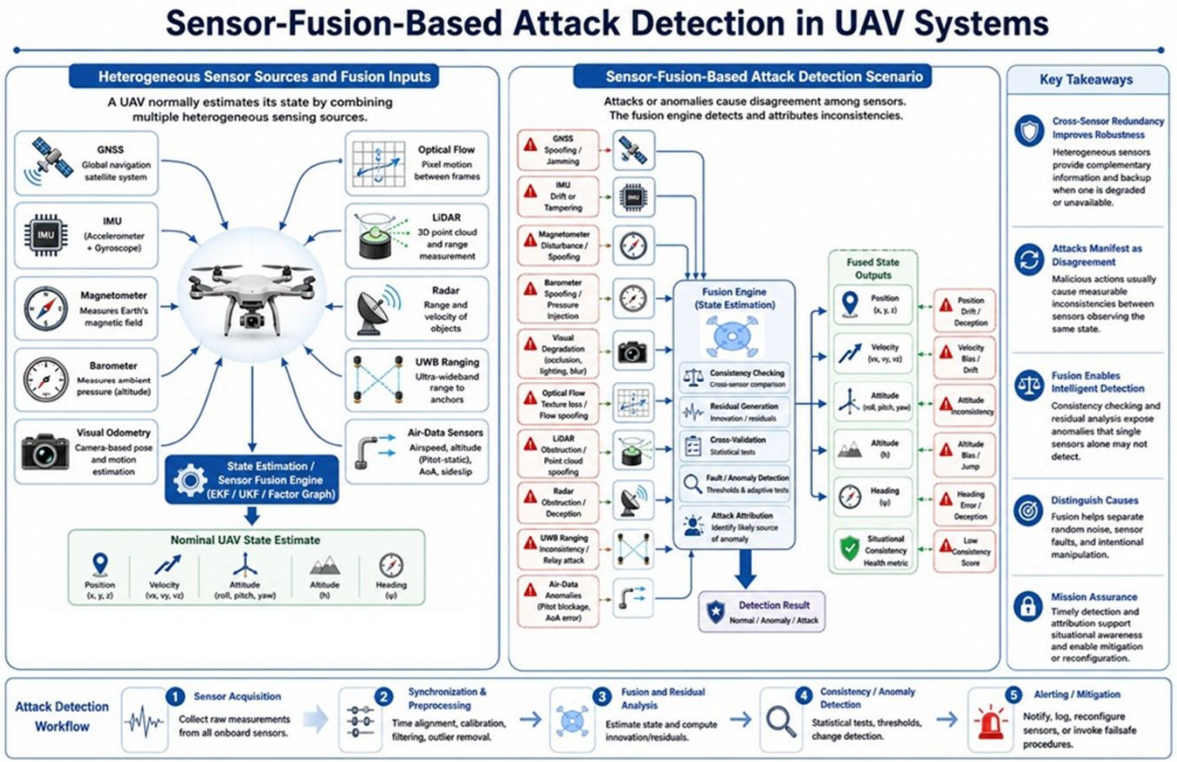


Figure S6.3. Cross-sensor residuals can reveal inconsistency when timing and uncertainty are modeled. Correlated errors or shared failures can reduce the value of redundant sensors.

Conceptual overview. Listed methods and outcomes do not establish measured detection performance.

Detection performance and operational cost

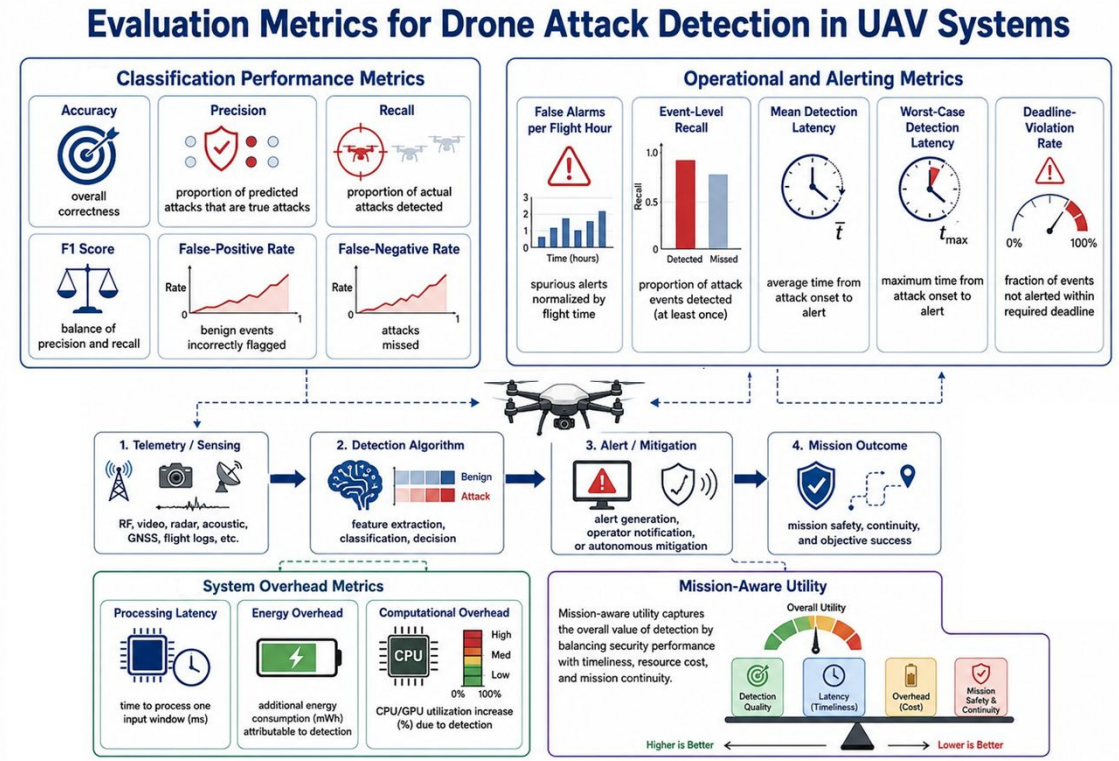


Figure S6.4. Evaluation should combine detection quality, alert timing, false alarms, and resource use. State event definitions, denominators, test conditions, and the treatment of missed detections.

Conceptual overview. Listed methods and outcomes do not establish measured detection performance.

Risk record: stale Wi-Fi drone commands

Illustrative Wi-Fi drone case: delayed control messages remain in the queue after their useful deadline.

Risk-record field	Illustrative entry	Evidence / review needed
Threat or failure scenario	A delayed or replayed message reaches the command application	Generation and reception times, sequence IDs, and acceptance log
Likelihood assessment	Not estimated from the information in this example	Observed delays, losses, exposure, and a stated confidence rationale
Mission consequence	Stale setpoints or delayed recognition of a degraded C2 link	Vehicle state, command age, watchdog timing, and mode trace
Candidate controls	Authenticated messages, freshness checks, sequencing, and link watchdog	Delayed-message and outage tests with explicit acceptance criteria
Residual risk and owner	To be reviewed after controls are tested in the stated conditions	Remaining failures, accountable reviewer, and retest decision

Likelihood, consequence, and confidence should remain explicit throughout the assessment.

Worked example: Wi-Fi command freshness

Synthetic application timestamps. Assume a verified ±2 ms relative clock-error bound.

Quantity	Assumed value	Meaning
Command generation timestamp	10.000 s	Protected timestamp from the controller
Drone reception timestamp	10.085 s	Timestamp at the freshness check
Relative clock-error bound	±2 ms	Total uncertainty in the time difference
Application freshness limit	80 ms	Maximum permitted age for this example

Observed age = (10.085 − 10.000) s = 85 ms

True command age lies between 83 ms and 87 ms

Minimum age = 83 ms > 80 ms freshness limit

Reject this stale command. The timing result alone does not establish an attack.

Detector evaluation and reporting protocol

Freeze the detector and its thresholds before final testing. Keep test flights separate from development data.

Evaluation area	Record and report	Pitfall to avoid
Data and generalization	Split by flight or session, record conditions, labels, and model version	Overlapping windows from one flight in both training and testing
Detection and false alarms	Event recall, precision, false alarms per benign flight hour, and counts	High accuracy dominated by normal data or duplicate alerts
Timing and mission behavior	Event onset, alert time, misses, deadline failures, and state changes	Reporting only processing time or ignoring missed detections
Compute and energy	Target hardware, load, incremental power, energy, and baseline	Comparing unmatched workloads or confusing power with energy

Assess expected maneuvers and unseen conditions, then report uncertainty and the limits of the test set.

Practical exercise: monitoring and risk review

Use a repeatable simulator or recorded dataset with known labels, independent reference data, and a benign baseline.

1. Baseline and hypothesis	Select one monitored quantity. Record normal variation, data quality, and the mission consequence.
2. Controlled comparison	Compare a benign maneuver, a sensor fault, and a simulated stale-data or navigation anomaly.
3. Detection evaluation	Measure residuals, alerts, missed events, false alarms, time-to-alert, and resource overhead.
4. Risk and response review	Explain what the detector establishes. Validate a bounded fallback and record remaining uncertainty.

Deliverable: a labeled event log, detector report, and risk record with a justified response policy.

References and terminology

Primary references support risk analysis, AI evaluation, and continuous-monitoring principles.

NIST SP 800-30r1 Guide for Conducting Risk Assessments (2012).
csrc.nist.gov/pubs/sp/800/30/r1/final

NIST AI 100-1 Artificial Intelligence Risk Management Framework, AI RMF 1.0 (2023).
www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10

NIST AIRC AI RMF Core. MEASURE function and evaluation documentation.
airc.nist.gov/airmf-resources/airmf/5-sec-core/

NIST SP 800-137 Information Security Continuous Monitoring (2011).
csrc.nist.gov/pubs/sp/800/137/final

GNSS: global navigation satellite system IMU: inertial measurement unit UWB: ultra-wideband
Residual: observation minus prediction W: power Wh: energy